



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

## Executive Summary

Fairness Experiment Report - Upload 18 summarizes the current fairness state for upload 18. Accuracy changed by +0.923 and fairness score changed by +0.943. Strong fairness improvement with no measurable accuracy loss. The best balanced model is Pipeline\_mitigated\_smote with combined score 0.935.

| ACCURACY          | FAIRNESS SCORE    | COMBINED SCORE                    | DISPARATE IMPACT     |
|-------------------|-------------------|-----------------------------------|----------------------|
| 0.923<br>▲ +0.923 | 0.943<br>▲ +0.944 | 0.935<br>Pipeline_mitigated_smote | 0.727<br>Ratio (DIR) |

## Dataset & Model Information

| DATASET               | MODEL                  | SECTIONS INCLUDED |
|-----------------------|------------------------|-------------------|
| processed_dataset.csv | german_dt_baseline.pkl | comparison        |

## Fairness Metrics

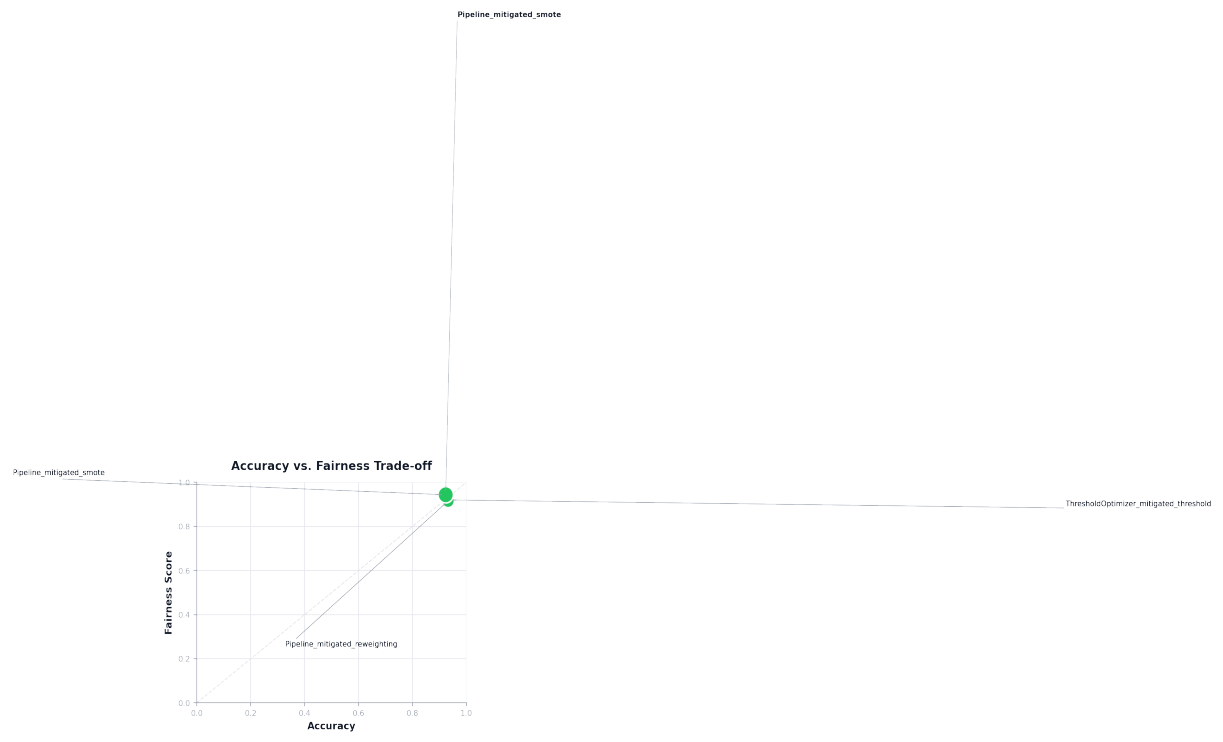
| METRIC | VALUE | INTERPRETATION |
|--------|-------|----------------|
|--------|-------|----------------|

|                                     |       |                                                                                           |
|-------------------------------------|-------|-------------------------------------------------------------------------------------------|
| Demographic Parity Difference (DPD) | 0.101 | Difference in positive-prediction rates across groups; closer to 0 is more equitable.     |
| Equalized Odds Difference (EOD)     | 0.012 | Gap in true/false positive rates between groups; closer to 0 indicates equal error rates. |
| Disparate Impact Ratio (DIR)        | 0.727 | Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.         |

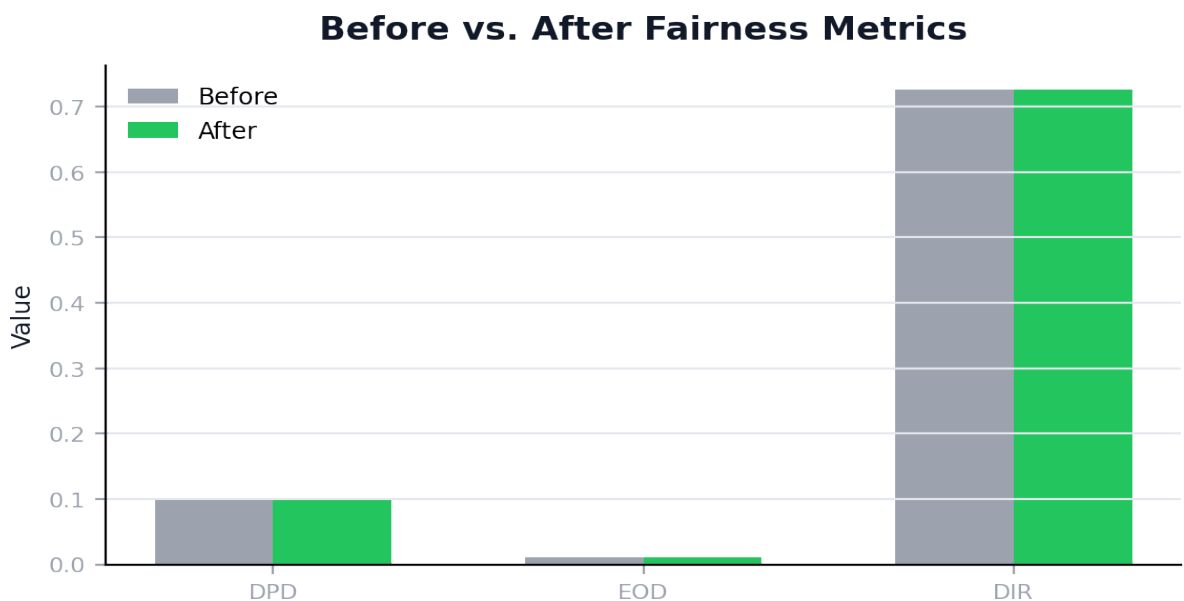
## Model Comparison

| Model                                  | Accuracy | Fairness | DPD   | EOD   | DIR   | Combined |
|----------------------------------------|----------|----------|-------|-------|-------|----------|
| ★ Pipeline_mitigated_smote             | 0.923    | 0.943    | 0.101 | 0.012 | 0.727 | 0.935    |
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| ★ Pipeline_mitigated_smote             | 0.923    | 0.943    | 0.101 | 0.012 | 0.727 | 0.935    |
| ThresholdOptimizer_mitigated_threshold | 0.935    | 0.920    | 0.099 | 0.061 | 0.734 | 0.926    |
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| ThresholdOptimizer_mitigated_threshold | 0.935    | 0.920    | 0.099 | 0.061 | 0.734 | 0.926    |
| Pipeline_mitigated_reweighting         | 0.932    | 0.915    | 0.100 | 0.069 | 0.722 | 0.922    |
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| Pipeline_mitigated_reweighting         | 0.932    | 0.915    | 0.100 | 0.069 | 0.722 | 0.922    |

## Accuracy vs. Fairness Trade-off



## Before vs. After Fairness Metrics



## Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.